

Loan Amortization Project

Purchasing a Home

TJ Collamore - Geometry Block G

Wish list or Goal list

- Nature View
- Academic School
- District Nearby
- 4 Bedrooms
- Safe Town
- Family-friendly
Neighborhood
- Large Yard
- Modern
- Central Air
- Garage

Possibilities



31 Green Ln,
Weston, MA 02493



9548 Bent
Grass Run,
Clarence
Center, NY
14032



2127 White Cove
Dr, La Marque,
TX 77568

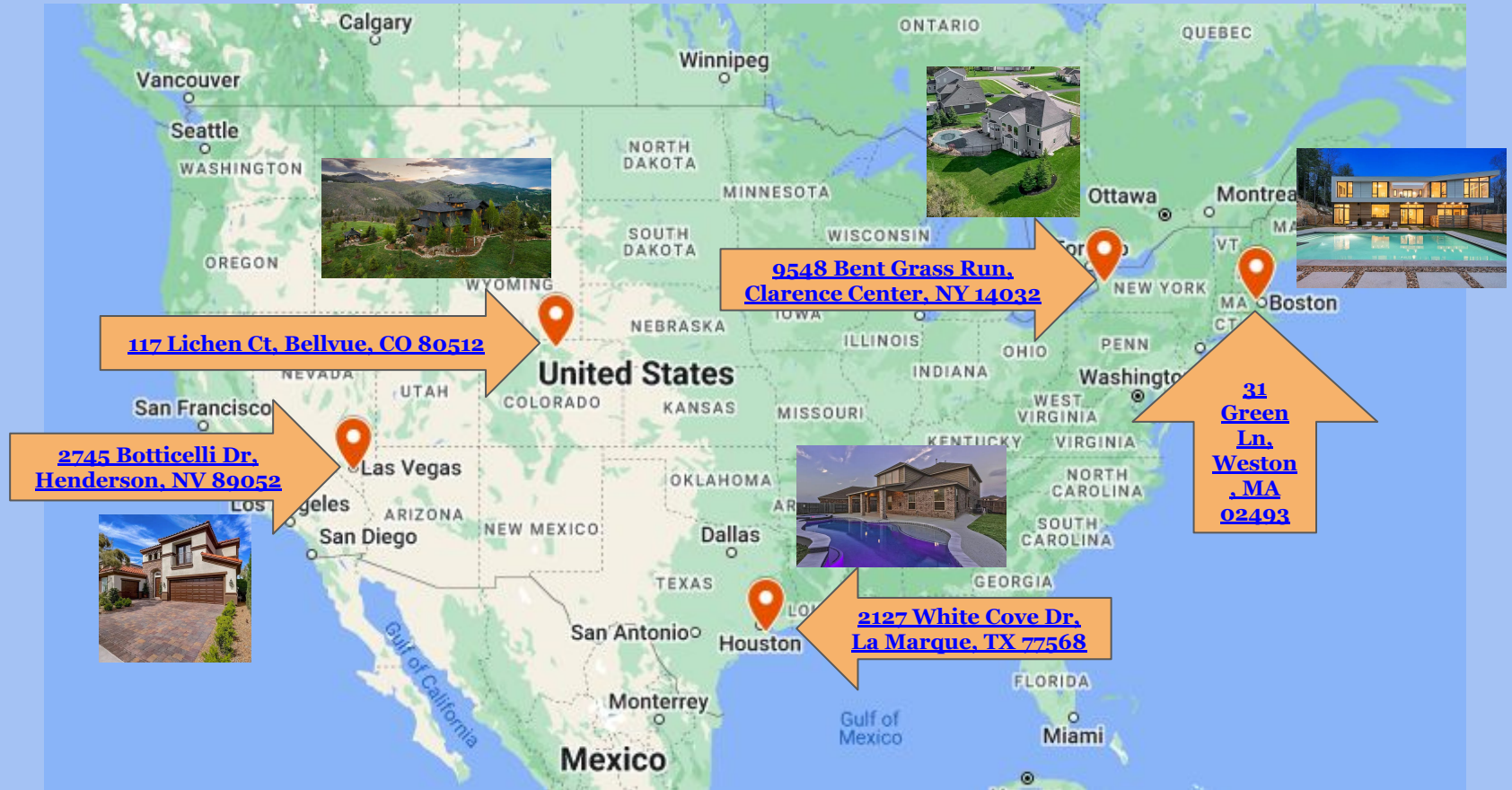


117 Lichen
Ct, Bellvue,
CO 80512



2745
Botticelli Dr,
Henderson,
NV 89052

Map



[Google Maps Link](#)

Evaluation of choices

Home Address	<u>31 Green Ln, Weston, MA 02493</u>	<u>9548 Bent Grass Run, Clarence Center, NY 14032</u>	<u>2127 White Cove Dr, La Marque, TX 77568</u>	<u>2745 Botticelli Dr, Henderson, NV 89052</u>	<u>117 Lichen Ct, Bellvue, CO 80512</u>
Schools	A+	B+	D	A-	B+
4 Bedrooms	A-	A+	A+	B	A+
Safe Town	B+	A	D-	B+	A
Neighborhood	B	A	A	A-	C
Large Yard	A-	B-	C-	B-	A+
Modern	A+	B	A-	C+	B+
Central Air	A+	A+	B+	A+	A+
Garage	A	B	B-	B	A+
Nature View	B	B-	C-	C	A+

Loan information

Bank of America - 6.875% for 30 yrs | 6.000% for 15 yrs

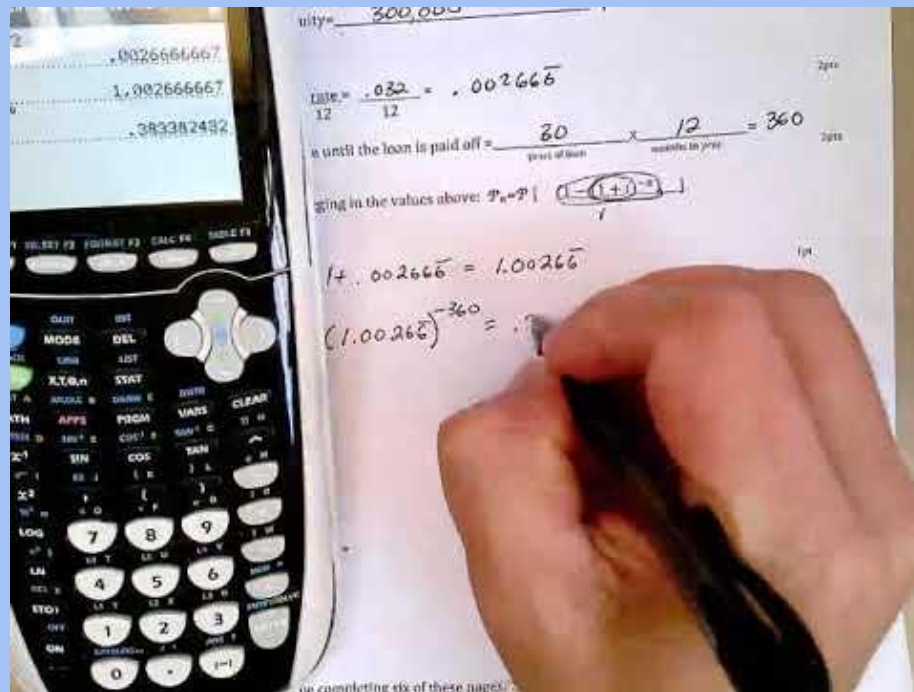
Rocket Mortgage - 6.375% for 30 yrs | 6.25% for 15 yrs

First Tech Federal Credit Union - 6.875% for 30 yrs | 6.375% for 15 yrs

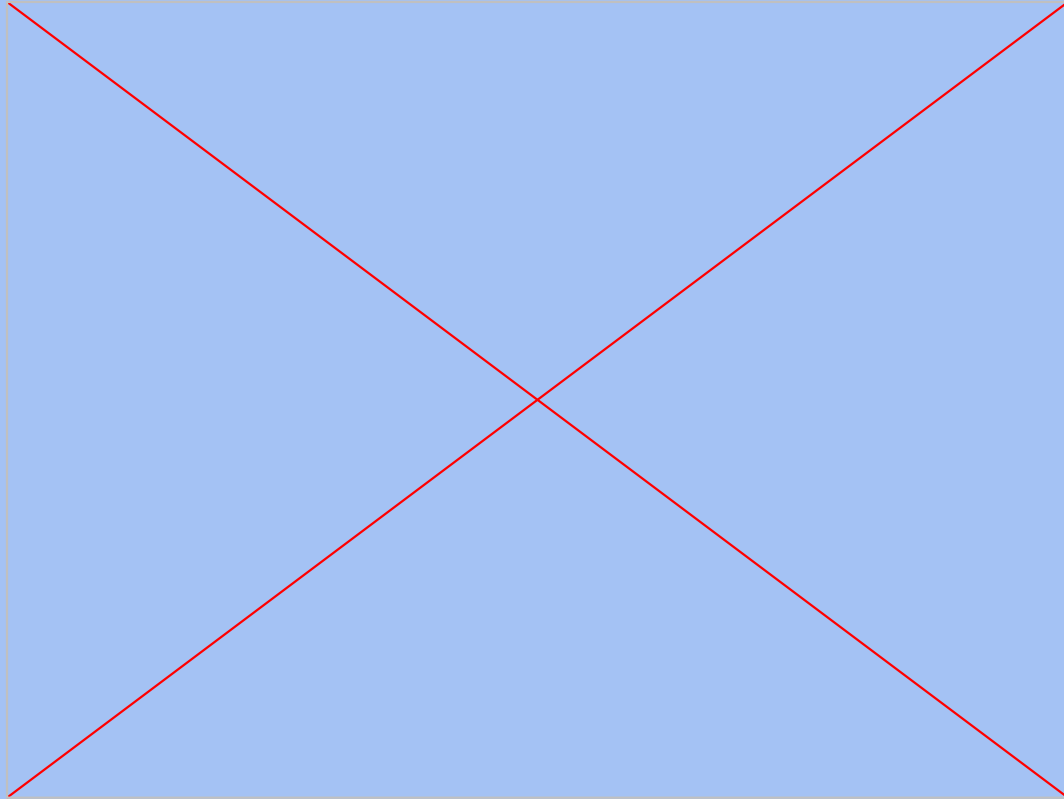
Citizens Bank - 6.75% for 30 yrs | 6.125% for 15 yrs

Chase - 6.49% for 30 yrs | 6.125% for 15 yrs

How to Complete Calculations



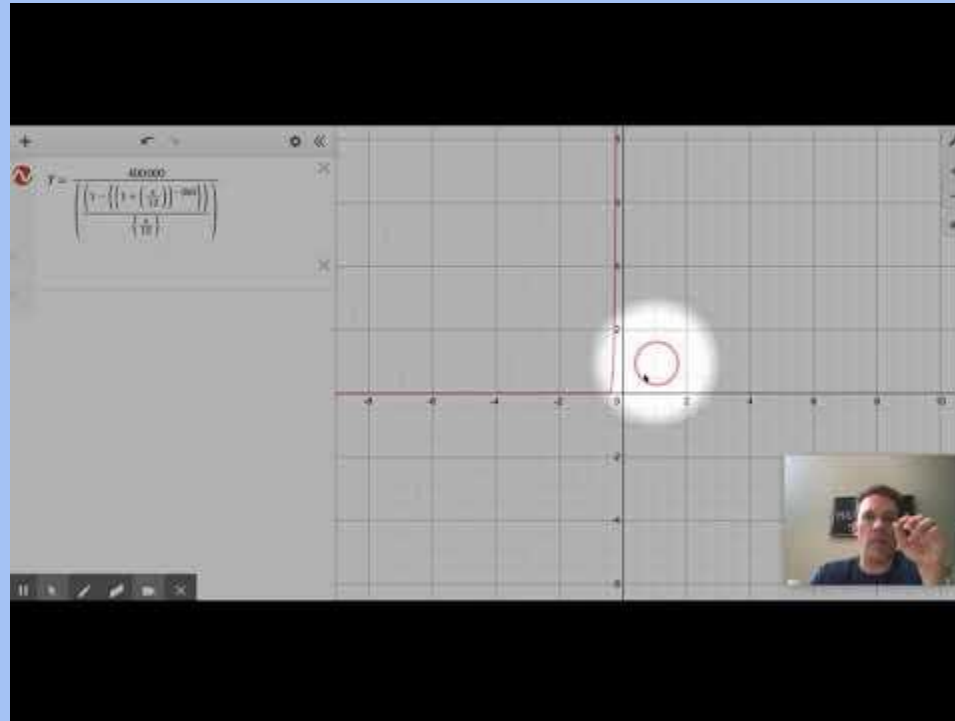
Video of Myself Completing Calculations



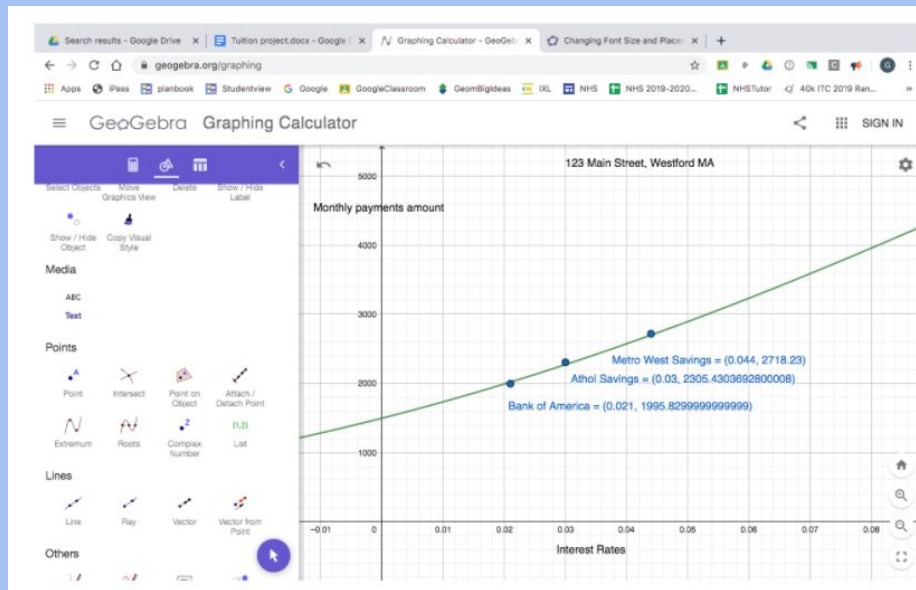
Linked PDFs Showing Calculations

House 1	House 2	House 3
<u>31 Green Ln, Weston, MA 02493 - 30 year</u>	<u>2127 White Cove Dr, La Marque, TX 77568 - 30 year</u>	<u>117 Lichen Ct, Bellvue, CO 80512 - 30 year</u>
<u>31 Green Ln, Weston, MA 02493 - 15 year</u>	<u>2127 White Cove Dr, La Marque, TX 77568 - 15 year</u>	<u>117 Lichen Ct, Bellvue, CO 80512 - 15 year</u>

Desmos Graph Tutorial

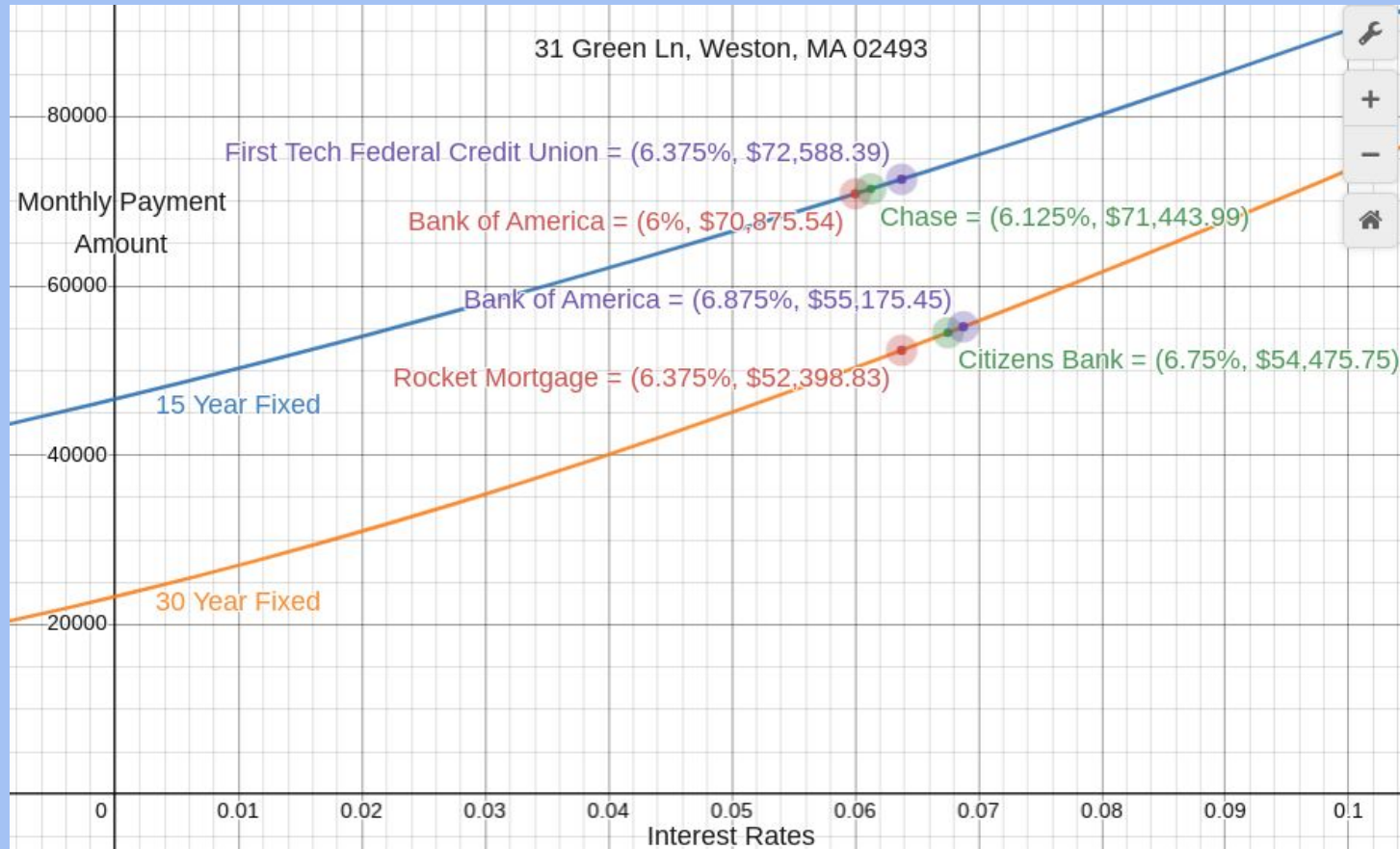


Desmos Instructions

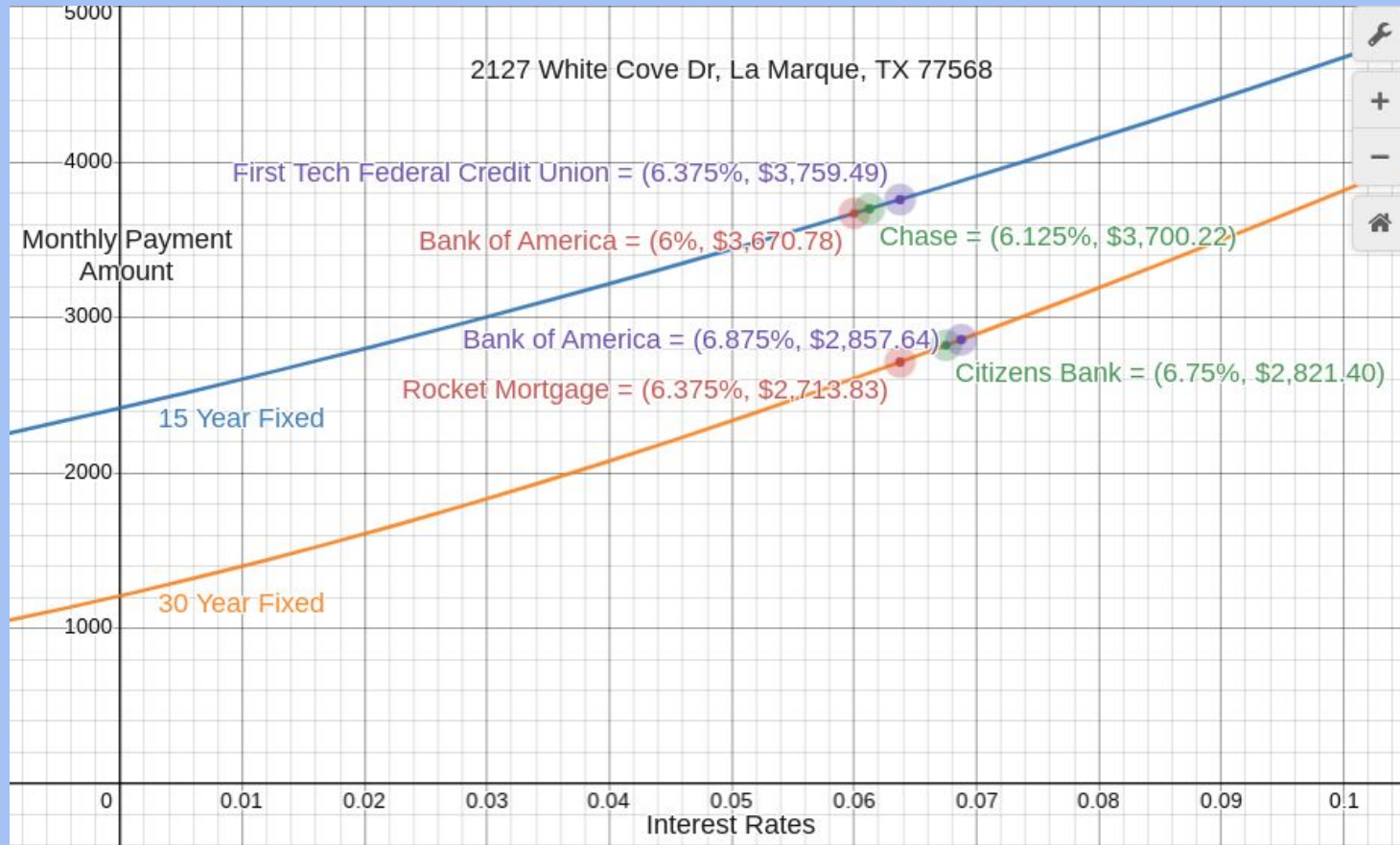


Steps To Use Desmos Link

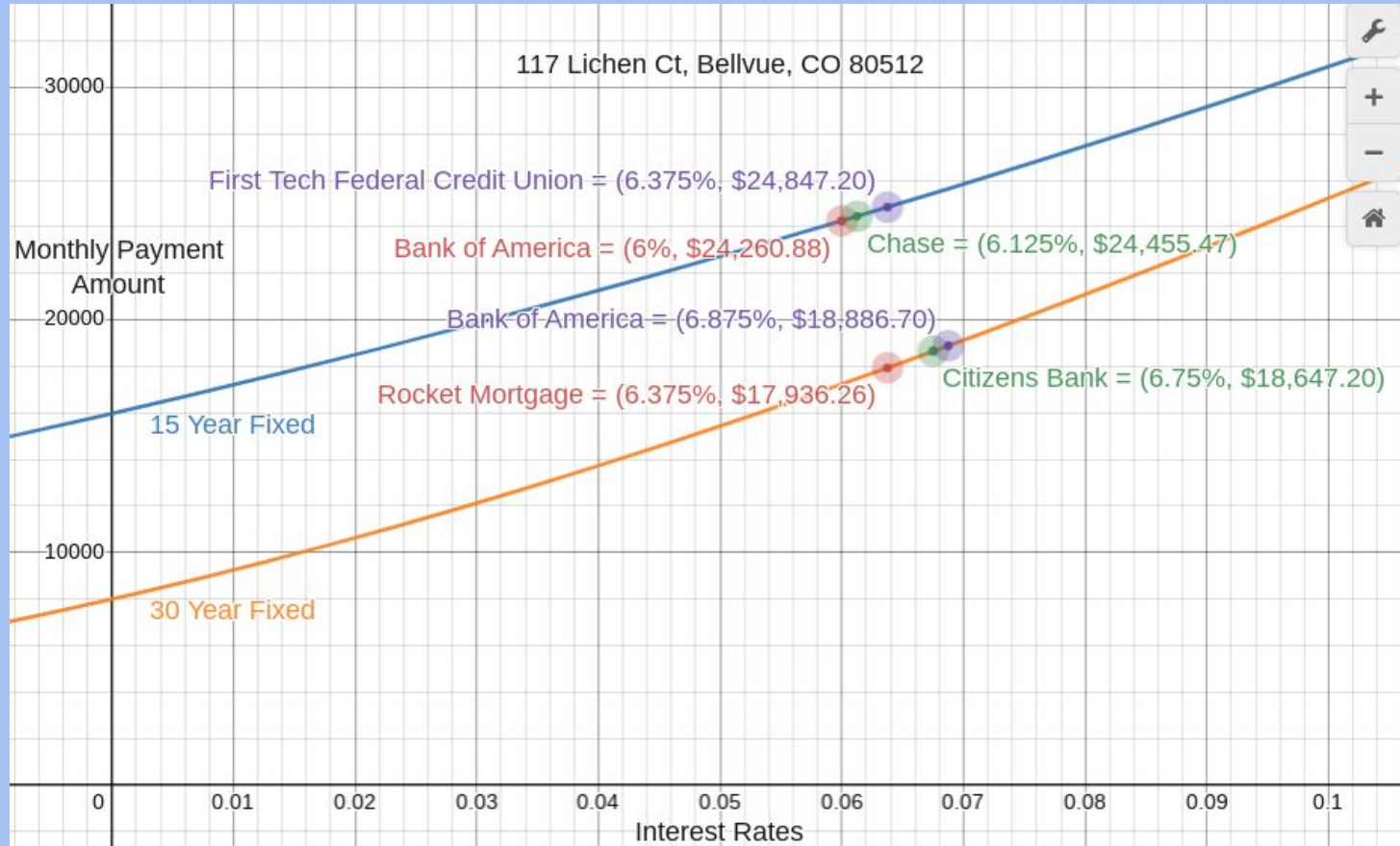
Graph #1



Graph #2



Graph #3



Loan and Home Ownership Survey

[Google Form Link](#)

Survey Results

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Name	1. What are the pros and cons between a 15 year loan and a 30 year loan? When would you use each one?	2. Is it better to take a loan or pay out of pocket to buy a house?	3. What were your compromises you had to make when choosing a house?	4. What wouldn't you want to sacrifice when choosing a house?	5. Are the formulas in the slideshow representative of actual mortgages?	6. What happens when you can't pay for a loan?	7. What happens if you sell your house before you have finished paying off your loans?	8. Does it actually take 30 or 15 years to pay off the loan?	9. Are home maintenance payments expensive relative to the mortgage payment?	10. When did you learn how to calculate loans? Who or where did you learn it from?
Brian Collamore	A 15 year loan is better if you can afford it, unless the interest rates are so low that you can put your money to work elsewhere.	If the interest rates are very low, it is better to take a loan. However, you should only do so if you are comfortable with having debt.	The house we got was smaller than we wanted (we originally hoped for four bedrooms), and it had a sump pump, which we feared would add anxiety (which it did).	I wouldn't want to sacrifice a good school district.	Kind of. The actual situation is more complicated. For example, real estate taxes are a significant portion of the monthly payment, but these were not included in the slides. Also, jumbo loans have higher interest rates. Also, if you don't have 20% down, you have to pay PMI. To avoid PMI, we took out two loans, the second being an adjustable rate HELOC, and then paid it off before interest rates increased a few years later. We also refinanced a couple times to reduce the interest rate.	The worst case scenario is called foreclosure, where the bank takes the house back, and you're kicked out.	You collect almost the full sale price, pay off the loan with those funds, and then the remainder of the funds can be used for your next home.	You can typically pay it off sooner. For example, by making one extra monthly payment per year, you can take about eight years off the 30 year mortgage. On the flip side, a refinance could reset the 30-year clock, unless for instance, you change from a 30 to a 15 year loan.	Yes. Maintenance costs are deceptively expensive. For example, a new furnace, roof, or windows can each run well over \$10,000 each.	I learned it when it was time for me to buy a house, just as the internet was becoming widely available.

Survey Results

2/3

Name	1. What are the pros and cons between a 15 year loan and a 30 year loan? When would you use each one?	2. Is it better to take a loan or pay out of pocket to buy a house?	3. What were your compromises you had to make when choosing a house?	4. What wouldn't you want to sacrifice when choosing a house?	5. Are the formulas in the slideshow representative of actual mortgages?	6. What happens when you can't pay for a loan?	7. What happens if you sell your house before you have finished paying off your loans?	8. Does it actually take 30 or 15 years to pay off the loan?	9. Are home maintenance payments expensive relative to the mortgage payment?	10. When did you learn how to calculate loans? Who or where did you learn it from?
Sara Moriarty	15 year- overall you pay less interest, however the monthly payments are higher. 30 year- overall you pay more interest, however the monthly payments are lower. If the interest rate is low, it would be beneficial to take a 30 year loan, and you would be able to afford a more expensive house. If the interest rate it high, it would be beneficial to take a 15 year loan, yet you would be able to afford a less expensive house to keep your monthly payments reasonable.	Take a loan	Size, # of bedrooms, # of bathrooms, closet size, age of bathroom	Neighborhood- good school system, house that needs a lot of work at the time of purchase	Yes, however, in this example, it looks as if the buyer is not making a downpayment. The amount of the downpayment will affect the monthly payments. It also does not consider that interest rates fluctuate daily and you need to "lock" in the interest rate to make the precise calculation for the monthly payment.	The bank takes the house back (foreclosure) and you file for bankruptcy	You owe the bank the remainder of the mortgage. This usually comes out of the profit you made on the house.	It depends- you can pay off the balance over the 15/30 years. You can make extra payments to take time off the back end, or refinance for a lower interest rate, and add time on the back end.	Depends on the month. Usually monthly mortgage payments are higher than home maintenance payments, however, throughout the entire 15/30 years, you usually incur some larger maintenance costs as the home is aging/needs upkeep.	When I was purchasing my first home. The mortgage broker educated me on the loan calculations.

Survey Results

3/3

Name	1. What are the pros and cons between a 15 year loan and a 30 year loan? When would you use each one?	2. Is it better to take a loan or pay out of pocket to buy a house?	3. What were your compromises you had to make when choosing a house?	4. What wouldn't you want to sacrifice when choosing a house?	5. Are the formulas in the slideshow representative of actual mortgages?	6. What happens when you can't pay for a loan?	7. What happens if you sell your house before you have finished paying off your loans?	8. Does it actually take 30 or 15 years to pay off the loan?	9. Are home maintenance payments expensive relative to the mortgage payment?	10. When did you learn how to calculate loans? Who or where did you learn it from?
Jason Moriarty	For a 15 year loan you will pay less in overall interest however your monthly payments will be higher. For a 30 year loan you will more in interest over the life of the loan however your monthly payments will be lower and manageable. If you have more disposable cash on hand it will be beneficial to consider a 15 year loan. Newer homeowners without much a portfolio will be more likely to choose a 30 year loan	If you have the cash it would be beneficial to use it to buy a house as it will reduce your over interest payments	It came down to price. So I had to choose a place that was basic	Location!	Yes if there is no down payment	Your home can be seized by the bank that originally approved you for a loan.	You would use the sales price of the home to put towards your remaining balance due. If the sales price didn't cover the remaining balance then you would owe the difference. If the sales price is higher then you can pocket the difference.	If you are paying the required amount each month the loan will most likely take the exact time of the loan.	Depending of the circumstances the maintenance can be very high compared to the mortgage payment.	Learned a lot in the home buying process from my mortgage officer.

Read and Reflection

Home Ownership Articles: [Article #1](#) [Article #2](#)

[Link to Google Document Reflection](#)